

ELECTRONIC NOSE-BASED DETECTION OF ONION FRESHNESS USING MOS SENSOR ARRAY AND MACHINE LEARNING CLASSIFICATION

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by

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CERTIFICATE

This is to certify that the project titled ELECTRONIC NOSE-BASED DETECTION OF ONION FRESHNESS USING MOS SENSOR ARRAY AND MACHINE LEARNING CLASSIFICATION has been submitted by Madhuhrita Saha under the supervision and guidance of Prof. Shubhajit Roy Chowdhury as a part of the requirements for the completion of SUMMER INTERNSHIP 2026 at the Centre for Human-Computer Interaction, Indian Institute of Technology Mandi.

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ABSTRACT

Postharvest spoilage of onions (*Allium cepa* L.) poses a significant challenge to food security and supply-chain efficiency, particularly in regions with inadequate cold-chain infrastructure, while early-stage fungal infection often evades reliable detection through conventional visual inspection. This study presents a low-cost, standalone electronic nose (E-nose) system for non-destructive, multi-class onion freshness classification, built around an ESP32 microcontroller interfaced with a five-sensor metal-oxide-semiconductor (MOS) gas array (MQ135, MQ137, TGS2600, TGS2602, TGS2620) and a DHT11 temperature-humidity sensor, housed in a custom-fabricated acrylic chamber assembly. The device executes a fully automated four-stage measurement protocol—chamber evacuation, manual valve-controlled headspace admission, synchronised 10 Hz sensor acquisition, and on-device inference—accessible through a self-hosted Wi-Fi web dashboard requiring no external computing infrastructure during routine operation. Training data comprising 60 observations across four freshness classes (Fresh, Slightly Aged, Moderately Spoiled, Highly Spoiled) were used to develop an offline MATLAB analysis pipeline incorporating Principal Component Analysis, exploratory Random Forest benchmarking, and a constrained, sensor-only decision tree subsequently deployed to the device via a JSON-based update mechanism. The deployed classifier achieved a cross-validated accuracy of 65.0% (macro-F1 = 0.542), with feature-importance analysis revealing a day-correlated environmental confound in the exploratory benchmark and an unanticipated reliance on only two of the five sensor channels. The complete prototype was realised at a hardware cost of approximately ₹5,960, demonstrating a practical, economically viable approach to real-time, edge-deployable postharvest quality monitoring.

Keywords: Electronic nose; MOS gas sensor array; onion freshness classification; ESP32 edge deployment; decision tree

INTRODUCTION

Vegetables are an integral and irreplaceable component of the human diet, serving as the primary sources of vitamins, minerals, dietary fibre, and antioxidants that support metabolic function, immune defence, and the prevention of non-communicable diseases [1]. Among the world's most widely produced and consumed bulb vegetables, the onion (*Allium cepa* L.) occupies a position of particular global and national significance. India is the second-largest producer and one of the largest exporters of onions in the world, with annual production exceeding 25 million tonnes, making the crop central to both domestic food security and agricultural export revenue. Onions are valued not only for their culinary versatility but also for their well-documented pharmacological properties, including antimicrobial, anti-inflammatory, and cardioprotective effects, attributed primarily to their rich organosulfur compound and flavonoid content.

However, the quality of onions and other vegetables is inherently dynamic and begins to degrade progressively from the moment of harvest. Degradation is driven by metabolic activity, enzymatic reactions, microbial colonisation, and environmental stressors such as humidity and temperature fluctuations during post-harvest handling, transportation, and storage [2]. As the interval between harvest and consumption increases, the nutritional value and sensory attributes of the vegetable deteriorate, ultimately rendering it unfit for human consumption. In India, where cold-chain infrastructure remains inadequate in large parts of the agricultural supply chain — particularly in hill regions such as Himachal Pradesh — post-harvest losses in vegetables routinely reach 20–40% of total production, threatening both household food security and the livelihoods of smallholder farmers [3].

Beyond economic considerations, the consumption of spoiled or compromised vegetables carries direct and serious health consequences. The World Health Organization estimates that approximately 600 million people suffer from foodborne illness annually, with associated mortality reaching hundreds of thousands of deaths each year [4]. In the context of onion freshness, a particular hazard is posed by fungal infection and mycotoxin contamination. Stored onions are susceptible to a range of pathogenic fungi, including species of *Fusarium*, *Aspergillus*, *Botrytis*, *Penicillium*, and *Alternaria*, which can penetrate the outer protective layers of the bulb and colonise inner tissue under conditions of elevated humidity, mechanical damage, or temperature variation [5]. The insidious nature of early-stage fungal infection — which often produces no visible discolouration or structural deformation before significant internal colonisation has occurred — makes detection through conventional visual inspection unreliable and subjective.

Traditional approaches to vegetable quality assessment range from empirical sensory evaluation to sophisticated analytical laboratory techniques such as gas chromatography-mass spectrometry (GC-MS) and high-performance liquid chromatography (HPLC). While highly sensitive and specific, these methods are time-consuming, require complex sample preparation, demand expensive instrumentation, and are ill-suited to real-time or on-site quality monitoring in agricultural settings [6]. This gap has driven intensive research interest toward sensor-based artificial intelligence solutions.

The electronic nose (E-nose) has emerged as one of the most promising non-destructive technologies for food quality assessment, drawing inspiration from the biological olfactory system. It is an intelligent sensing platform that integrates an array of chemical gas sensors exhibiting partial cross-sensitivity to a range of volatile compounds, a signal conditioning and data acquisition unit, and a pattern recognition engine employing machine learning algorithms to classify and interpret the composite sensor response patterns [7]. Metal oxide semiconductor (MOS) sensors, owing to their broad chemical sensitivity, low cost, and compatibility with miniaturised hardware platforms, constitute the most widely deployed sensor type in food-grade E-nose systems [8]. Studies have demonstrated the capacity of E-nose systems equipped with MOS sensor arrays to detect early-stage spoilage in potatoes and onions with classification accuracies exceeding 90% [9]. In the specific context of onion quality, the E-nose has shown sensitivity to the sulphur-containing volatile compounds characteristic of *Allium* tissue as well as to the altered VOC profiles associated with fungal infection and bacterial soft rot, facilitating the non-invasive differentiation of healthy from infected bulbs before any visible external symptom appears [5, 10].

The present work addresses these needs by developing a compact, low-cost E-nose prototype specifically designed for onion freshness detection. The system was built around an ESP32 microcontroller interfaced with a five-sensor MOS array and a DHT11 temperature and humidity sensor, housed in a custom-fabricated acrylic chamber assembly. A fully automated measurement protocol and real-time Wi-Fi dashboard were implemented to enable practical, user-accessible freshness assessment. The work was conducted over a two-month summer internship period at IIT Mandi, encompassing literature review, hardware design, firmware development, data acquisition, and preliminary analysis, with machine learning classification planned as the next phase.

LITERATURE SURVEY

The detection of postharvest disease and freshness deterioration in onions using volatile organic compound (VOC) analysis has been the subject of growing research interest over the past two decades. The following section reviews the most relevant prior work that informed the design of the present system.

2.1 Foundational Work on Gas Sensor Arrays for Onion Disease Detection

Li, Schmidt, and Gitaitis (2011) conducted one of the earliest systematic studies on the detection of onion postharvest diseases using a gas sensor array paired with GC-MS analysis [11]. The study inoculated onions with *Botrytis allii* (causal agent of neck rot) and *Burkholderia cepacia* (sour skin), measuring headspace volatiles from 5 to 11 days after inoculation. PCA score plots demonstrated clear separation between healthy and infected samples, and a linear discriminant model achieved classification rates exceeding 97.8% across three treatment groups. GC-MS analysis identified 24 major volatile compounds from the headspace, with 16 unique to the infected bulbs, establishing that VOC profiling could reliably distinguish infection states.

Building on this foundation, Konduru, Li, and Rains (2012) developed a customised five-sensor MOS array specifically for postharvest disease detection in onions during storage [12]. The device automated data logging through a microcontroller (PIC 16F877A) and used a specially designed Teflon sensor chamber with a gas delivery system. The study demonstrated differences in volatile type and concentration between healthy, sour-skin-infected, and *Botrytis* neck rot-infected onions, validating the applicability of a low-cost MOS array as a diagnostic tool.

Konduru, Rains, and Li (2015) subsequently published a detailed system development and characterisation study of a seven-sensor MOS array for onion quality evaluation [13]. The sensor array was characterised against four chemicals (acetone, acetonitrile, ethyl acetate, ethanol) at multiple concentrations, demonstrating the array's ability to differentiate chemical type and concentration. A proof-of-concept classification of sour skin-infected onions achieved 89% accuracy, confirming the discriminative power of a carefully selected MOS sensor array for onion quality work.

Complementing these onion-specific designs, low-cost MQ-series metal-oxide sensors have similarly been validated as effective freshness estimators across a range of fruits and vegetables, supporting their suitability as the basis for the present sensor array [14].

2.2 Classification Methods and Machine Learning for Onion Quality

Konduru, Rains, and Li (2015) carried out the most comprehensive classification study in the onion E-nose domain [15]. Three baseline correction methods (relative, differential, fractional) and three feature extraction approaches were systematically compared using MANOVA to identify the optimal combination. Both Linear Discriminant Analysis (LDA) and Support Vector Machine (SVM) with radial basis function (RBF) kernel were evaluated as classifiers. SVM consistently outperformed LDA; the best cross-validated accuracy exceeded 92%, and in the most rigorous validation scenario (training on one batch, testing on another), SVM achieved 85.26% compared to LDA's 81.58%. The study also found that the relative response feature combined with relative baseline correction provided the best discrimination, a finding directly relevant to the preprocessing pipeline planned for the present work.

Li et al. (2011) similarly employed PCA and LDA in their multi-disease detection study, finding that the three-class problem (healthy, Botrytis neck rot, sour skin) was solvable with high accuracy when sufficient post-inoculation time had elapsed for VOC profiles to diverge [11]. The study noted that at early infection stages (5 days after inoculation), cluster separation was already visible in PCA score plots, suggesting that VOC-based detection can precede visible symptoms by several days.

Beyond onion-specific classifiers, recent work has explored more expressive machine learning architectures for E-nose pattern recognition, including intelligent framework designs for perishable food assessment [16], knowledge-distillation-based approaches to compensate for sensor drift in gas recognition [17], and convolutional neural networks trained on combined time- and temperature-domain features for odour classification [18]. These advances illustrate a broader trend toward deep learning-based E-nose classifiers, though the constrained, human-auditable decision tree adopted in the present work was selected specifically for its compatibility with the ESP32's limited memory budget (Section 4.6).

2.3 Fungal Infection Detection in Onions by E-nose

Labanska et al. (2022) specifically evaluated the capacity of an E-nose to detect *Fusarium oxysporum* f. sp. *cepae* infection in onions — the fungal pathogen most directly relevant to the present study [5]. The work demonstrated that PCA plots could reveal progressive changes in VOC profiles corresponding to disease development over time, and that storage at low temperature inhibited infection development, altering the VOC signature accordingly. This study confirmed that fungal infection — as distinct from bacterial diseases — produces a distinctive and detectable headspace volatile fingerprint accessible to an E-nose, even at early infection stages.

2.4 Electronic Nose for Potato and Onion Spoilage — Indian Context

Ghosh et al. (2022) developed an E-nose system named e-POT for the early spoilage detection of both potato and onion during post-harvest storage, using a MOS sensor array with Principal Component Analysis (PCA) and Probabilistic Neural Network (PNN) classification [9]. The study, conducted partly at research institutions in India including C-DAC Kolkata and CIAE Bhopal, is particularly relevant to the present work given its Indian agricultural context and its

demonstration that E-nose technology can be practically deployed for the types of crops and storage conditions representative of Indian agricultural systems. The study reported encouraging discrimination results between fresh and spoiled samples across both crop types.

This Indian-context validation parallels a broader international trend of E-nose deployment across diverse food matrices, including chicken meat freshness monitoring under both ambient and refrigerated storage [19, 20], freshness detection in fish [21], and real-time refrigerated freshness assessment using miniaturised E-nose hardware [22], collectively reinforcing the cross-commodity applicability of the sensing approach adopted here.

2.5 Onion Headspace Volatiles and Environmental Factors

Abbey et al. (2004) investigated the effect of nitrogen, sulphur, and soil type on onion headspace volatiles and bulb quality as assessed by an E-nose [23]. This work established that the E-nose response was sensitive not only to the freshness state of the onion but also to agronomic and compositional variables, highlighting the importance of controlled experimental conditions during data collection. The findings underlined that factors such as sulphur content — which directly influences the concentration of thiosulphinate and sulphide volatile compounds — can confound classification models if not accounted for in sample preparation and experimental design, a consideration echoed in subsequent E-nose storage-monitoring studies that explicitly control for environmental variability [2].

2.6 Summary and Gap Analysis

The reviewed literature collectively establishes that: (i) MOS sensor arrays can reliably detect infection-associated VOC changes in onions, a finding corroborated across broader food-quality review literature [3, 6]; (ii) SVM and LDA are the most validated classifiers for binary and multi-class onion freshness discrimination, consistent with comparative reviews of sensor-array-based classification pipelines [1]; (iii) PCA provides both a visualisation and dimensionality reduction tool well suited to E-nose data, as reaffirmed in recent comprehensive surveys of E-nose signal processing [7]; and (iv) the relative response feature combined with relative baseline correction is currently the best-practice preprocessing approach for MOS sensor data in this domain, in line with sensor-array optimisation studies [8]. A notable gap in existing literature is the integration of E-nose hardware with real-time wireless data transmission and IoT-accessible user interfaces, which limits the practical deployability of laboratory prototypes [24, 25, 26]. The present work directly addresses this gap through ESP32-based Wi-Fi connectivity and a web dashboard.

EXPERIMENTAL SETUP

3.1 Overview of the System Architecture

The developed E-nose system is composed of three integrated subsystems: (i) a custom-fabricated sensing and sample chamber assembly, (ii) an electronic hardware platform comprising the sensor array and microcontroller, and (iii) a firmware and web-interface layer that manages automated measurement cycles and real-time data presentation. A schematic overview of the system architecture and measurement workflow is described in the sections below.

3.2 Chamber Design and Fabrication

Both the sensor chamber and sample chamber were fabricated in-house at IIT Mandi using 5 mm cast acrylic (PMMA) sheets cut and bonded with acrylic solvent adhesive. Acrylic was selected for its chemical inertness to the sulfur-containing volatile compounds emitted by onions, its transparency (allowing visual inspection of the sample), and its machinability with available laboratory tools.

The sensor chamber, housing all MOS sensors and the DHT11 probe, has internal dimensions of 150 mm $\times$ 100 mm $\times$ 75 mm (L $\times$ W $\times$ H), giving an internal headspace volume of approximately 1,125 cm³. Ports were drilled and fitted with push-fit pneumatic connectors for inlet and outlet gas connections.

The sample chamber, in which the onion sample is placed for headspace accumulation prior to analysis, has internal dimensions of 75 mm $\times$ 75 mm $\times$ 87 mm (L $\times$ W $\times$ H), yielding a headspace volume of approximately 490 cm³. The chamber lid is sealed with a thin strip of laboratory Parafilm prior to headspace accumulation to prevent volatile escape.

Gas flow between the sample chamber and sensor chamber is mediated by a Kaomer HDVP1-B12 vacuum pump (rated flow: 1,000 mL/min) connected via 4 mm outer-diameter PTFE (Teflon) tubing. A 2-way manual ball valve is installed between the sample chamber outlet and the pump inlet, allowing the operator to open or close the headspace pathway on command from the web interface. PTFE tubing was specifically chosen to minimise adsorption of sulfur-containing VOCs on the internal tubing walls, an issue reported in earlier gas delivery systems using silicone or PVC tubing.

3.3 Bill of Materials and Cost Estimate

The total cost of the prototype, calculated at Indian retail prices (June 2026), is summarised in Table 1 below.

Component	Qty	Unit Price (₹)	Total (₹)
ESP32 Development Board (38-pin)	1	450	450
MQ-135 Gas Sensor Module	1	150	150
MQ-137 Gas Sensor Module	1	200	200
TGS2600 Gas Sensor (Figaro)	1	650	650
TGS2602 Gas Sensor (Figaro)	1	750	750
TGS2620 Gas Sensor (Figaro)	1	700	700
DHT11 Temperature & Humidity Sensor	1	80	80
SSD1306 OLED Display (128×64, I2C)	1	200	200
Kaomer HDVP1-B12 Vacuum Pump (1000 mL/min)	1	1,200	1,200
2-Way Manual Ball Valve (4mm push-fit)	1	120	120
PTFE Tubing (4mm OD, per metre)	2	80	160
Acrylic Sheet (5mm, A3, cast)	2	250	500
PCB, connectors, resistors, capacitors, misc.	—	—	800
TOTAL ESTIMATED COST			₹5,960

Table 1: Bill of materials and cost estimate for the E-nose prototype.

This places the total hardware cost at approximately ₹5,960 (~\$72 USD), significantly below the \$300 (approximately ₹25,000) reported by Konduru et al. (2015) for a comparable customised MOS-based E-nose system [3], and orders of magnitude below the cost of commercial E-nose instruments, which typically range from ₹5–50 lakhs.

3.4 Sensor Array: Selection and Rationale

The five gas sensors were selected based on their complementary sensitivity profiles to the volatile compounds known to be emitted by onions during freshness degradation and fungal infection. Table 2 summarises the sensors and their primary target gases.

Sensor	Type	Primary Target Volatiles	Relevance to Onion
MQ-135	MOS (SnO ₂)	NH ₃ , benzene, CO ₂ , smoke	Bacterial metabolites, air quality index
MQ-137	MOS (SnO ₂)	Ammonia, sulfur compounds	Key onion VOC marker; sour skin indicator
TGS2600	MOS (SnO ₂)	General air contaminants, H ₂ , CH ₄	Broad-spectrum freshness baseline
TGS2602	MOS (SnO ₂)	VOCs, H ₂ S, NH ₃ , odorous gases	Spoilage/rot odour signature
TGS2620	MOS (SnO ₂)	Alcohol, organic solvents	Fermentation-related volatile changes
DHT11	Capacitive / Resistive	Temperature (0–50°C), Humidity (20–80% RH)	Chamber condition monitoring

Table 2: Sensor array composition and target volatile profile.

MOS sensors were chosen due to their practical advantages for this application: they are easy to operate, commercially available at low cost, sensitive to a wide range of volatile compounds, and compatible with the ADC inputs of the ESP32 microcontroller, which provides a 12-bit resolution (0–4095 ADC counts) at 3.3 V reference. All five MOS sensors output an analogue voltage proportional to the concentration of target gases in the ambient atmosphere; the ESP32 reads these via its built-in ADC on GPIO pins 34, 35, 32, 33, and 36 respectively. The DHT11 communicates over a single-wire digital protocol on GPIO pin 4.

3.5 Electronic Hardware and Microcontroller

The ESP32 microcontroller (Espressif Systems) was selected as the central processing unit due to its dual-core Xtensa LX6 processor (240 MHz), integrated 802.11 b/g/n Wi-Fi, built-in 12-bit ADC, I2C and UART peripherals, and 4 MB flash with SPIFFS file system support — all in a compact, breadboard-compatible development module available for under ₹450. The onboard SPIFFS file system is used to store the web dashboard HTML in firmware memory and to write experimental CSV data files during measurement runs.

An SSD1306 128×64 OLED display is connected to the I2C bus (SDA: GPIO 21, SCL: GPIO 22) to provide a local status readout of system state, sensor readings, and classification outcome during operation, without requiring the user to keep a browser open. The vacuum pump is driven via a PWM signal on GPIO 25 (LEDC channel), *allowing* speed control through the

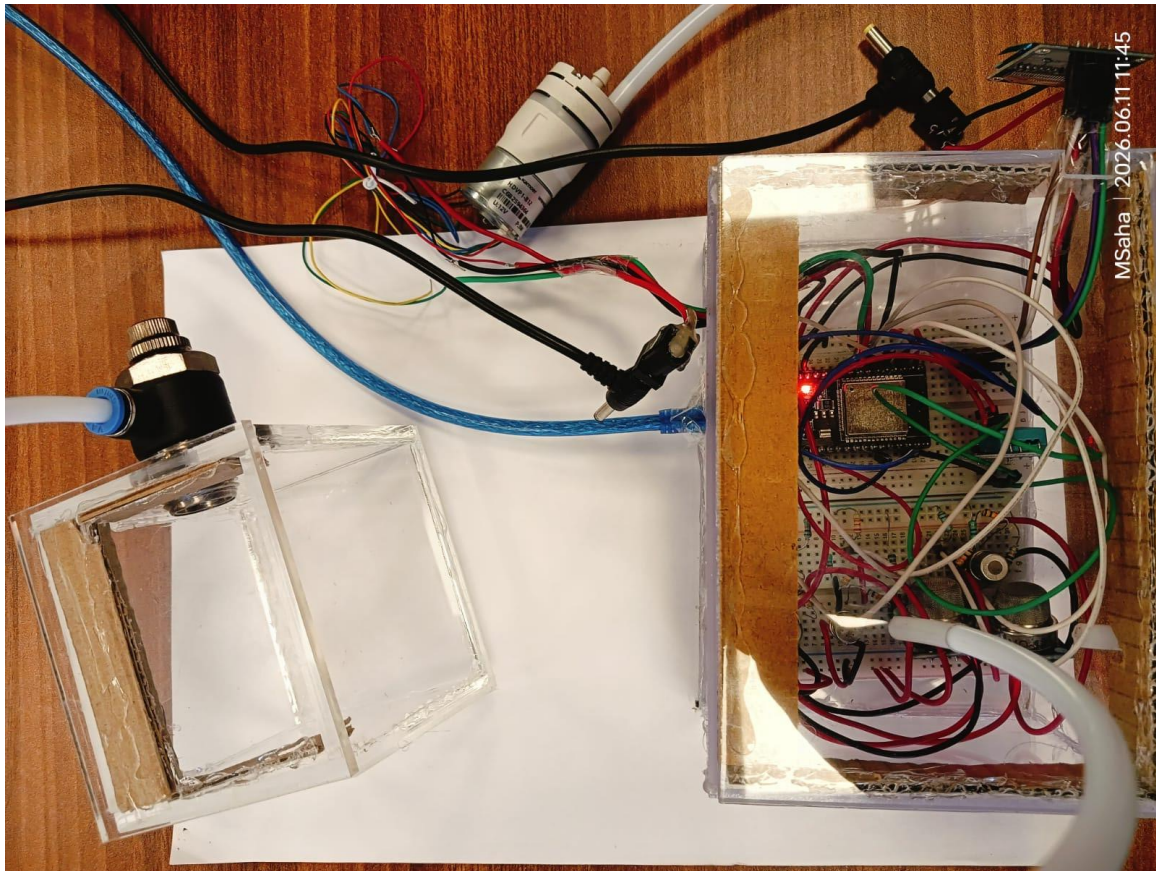


Figure 1. Experimental Setup

firmware. The pump is energised (PWM duty = 0, active-low driver) during the evacuation phase and de-energised (duty = 255) during sensing.

3.6 Sample Preparation

3.6.1 Onion Procurement and Initial Selection

Fresh onion bulbs (*Allium cepa* L., variety: Nasik Red, a widely cultivated short-day variety in peninsular India and commonly available in northern Indian markets) were procured from a local vegetable market in Mandi district, Himachal Pradesh. Bulbs were inspected at the point of purchase, and only those free of major bruising, visible surface wounds, or external fungal growth were selected. Both a 'fresh' cohort (intended for use as healthy/control samples) and a separate cohort (intended for accelerated spoilage induction) were purchased from the same vendor on the same day to minimise inter-batch variability in sulphur content and volatile baseline.

3.6.2 Surface Sterilisation and Pre-Treatment

To eliminate adventitious microbial contamination on the outer dry skin of the bulbs, which could confound headspace volatile measurements, the following pre-treatment protocol was followed for both sample groups:

1. The outermost dry papery skin layer was peeled off by hand.
2. The exposed bulb surface was wiped with a cotton swab soaked in 70% ethanol solution and allowed to stand for 10 minutes to effect surface sterilisation.
3. The bulbs were then rinsed with distilled water to remove chemical residues and patted dry with clean tissue paper.
4. Bulbs were placed on a clean laboratory bench at room temperature ($26 \pm 2^\circ\text{C}$, consistent with ambient laboratory conditions at IIT Mandi in the month of May–June) and allowed to dry for 1 hour before further handling.

3.6.3 Fresh (Control) Sample Preparation

The control group bulbs were prepared by injecting 1 mL of sterile distilled water into each bulb using a sterile hypodermic syringe to simulate the minimal tissue disruption introduced by the inoculation process applied to the infected group, ensuring that any mechanical handling effects were equivalent between the two groups. Control bulbs were then placed in individual sealed polypropylene containers and allowed to equilibrate at room temperature for 3 hours. Prior to each measurement session, a control bulb was placed in the sealed sample chamber and allowed to accumulate headspace volatiles for 6 hours before analysis.

3.6.4 Infected Sample Preparation and Spoilage Induction

The infected sample group was prepared to represent onions in an advanced state of fungal-associated spoilage, corresponding to the conditions encountered in poorly ventilated storage settings typical of hill-region agricultural warehouses. Spoilage induction was achieved through the following protocol:

5. Each bulb in the infected cohort was lightly scored on the outer surface using a sterile scalpel to create minor surface wounds, replicating the mechanical damage commonly occurring during harvest, handling, and transport.
6. The wounded bulbs were placed in a sealed incubation container maintained at approximately 30°C (achieved using a thermostatically controlled laboratory incubator available at IIT Mandi) for 48 hours. This temperature corresponds to the optimum growth range for *Fusarium* species and post-harvest bacterial pathogens commonly reported in onion storage literature [1, 4].
7. Following incubation, the infected bulbs were transferred to room temperature ($26 \pm 2^\circ\text{C}$) for 3 hours of conditioning before being individually sealed in the sample chamber.
8. As with the control group, each infected sample was allowed to accumulate headspace volatiles in the sealed sample chamber for 6 hours before the measurement protocol was initiated.

It should be noted that the present study did not involve inoculation with a specific isolated fungal or bacterial pathogen culture; rather, the accelerated spoilage protocol relies on the natural microflora present on the onion surface and in the incubation environment following mechanical wounding. This approach, while less controlled than pure-culture inoculation, is practical for a resource-constrained internship setting and produces a realistic simulation of the spoilage conditions encountered in actual post-harvest storage.

METHODOLOGY

4.1 System Overview

This work presents a hybrid edge-computing architecture for real-time, multi-class onion freshness classification using a low-cost electronic nose (e-nose) built on the ESP32 microcontroller platform. The system integrates five gas sensors with a microcontroller-hosted web interface, separating offline model development (performed in MATLAB) from on-device inference (performed entirely on the ESP32, with no continuous network or cloud dependency during normal operation).

The ESP32 hosts a web server at 192.168.4.1 through which an operator controls the full test cycle. Pressing **Start Test** on the dashboard triggers an automated sequence: the vacuum pump runs for 90 seconds to evacuate the sensing chamber, after which the system pauses for a 60-second window during which the operator manually opens the intake valve to admit headspace gas from the onion sample. Data acquisition then proceeds automatically for 60 seconds at 10 Hz across all five gas sensor channels plus temperature and humidity, after which results are classified and presented on the web dashboard, with the underlying CSV log available for download. A **Stop** button is available throughout the cycle to interrupt the routine at any stage.

Classification follows a two-stage flow layered on top of this acquisition cycle: the streaming sensor data collected during the 60-second acquisition window is first reduced, on-device, to a fixed-length feature vector per sensor channel (capturing response level, magnitude, and trend); these features are then passed through a lightweight decision-tree classifier — trained offline in MATLAB and deserialized on the ESP32 from a stored model file — to produce one of four freshness-class predictions, with no cloud or external compute required at inference time.

4.2 Hardware Architecture and Sensing Protocol

The sensing platform comprises an ESP32 microcontroller interfaced with five gas sensors (MQ135, MQ137, TGS2600, TGS2602, TGS2620) selected to capture complementary volatile organic compound (VOC) profiles associated with onion decomposition, a DHT11 sensor for chamber temperature and humidity, a PWM-driven vacuum pump for chamber evacuation, a manually operated intake valve, and an SSD1306 OLED display for local status readout. The ESP32 hosts both a Wi-Fi access point (192.168.4.1) and an asynchronous HTTP server, allowing any Wi-Fi-enabled device to control the system and view results without internet connectivity.

Each measurement cycle follows a fixed four-stage protocol, initiated via the "Start Test" control on the web dashboard:

- Evacuation (90 s) — the vacuum pump purges the sensing chamber of residual headspace gas.
- Valve-open window (60 s) — the operator manually opens the intake valve to admit headspace gas from the onion sample.
- Acquisition (60 s at 10 Hz) — all five gas sensor channels, plus temperature and humidity, are sampled synchronously and logged to onboard flash storage (SPIFFS) as a CSV file.
- Classification — feature extraction and inference are performed immediately following acquisition, with the result displayed on the dashboard and OLED.

Onion Freshness Classification

ESP32-Based Electronic Nose for Multi-Class Onion Freshness Classification: A Standalone Edge-Deployment Pipeline

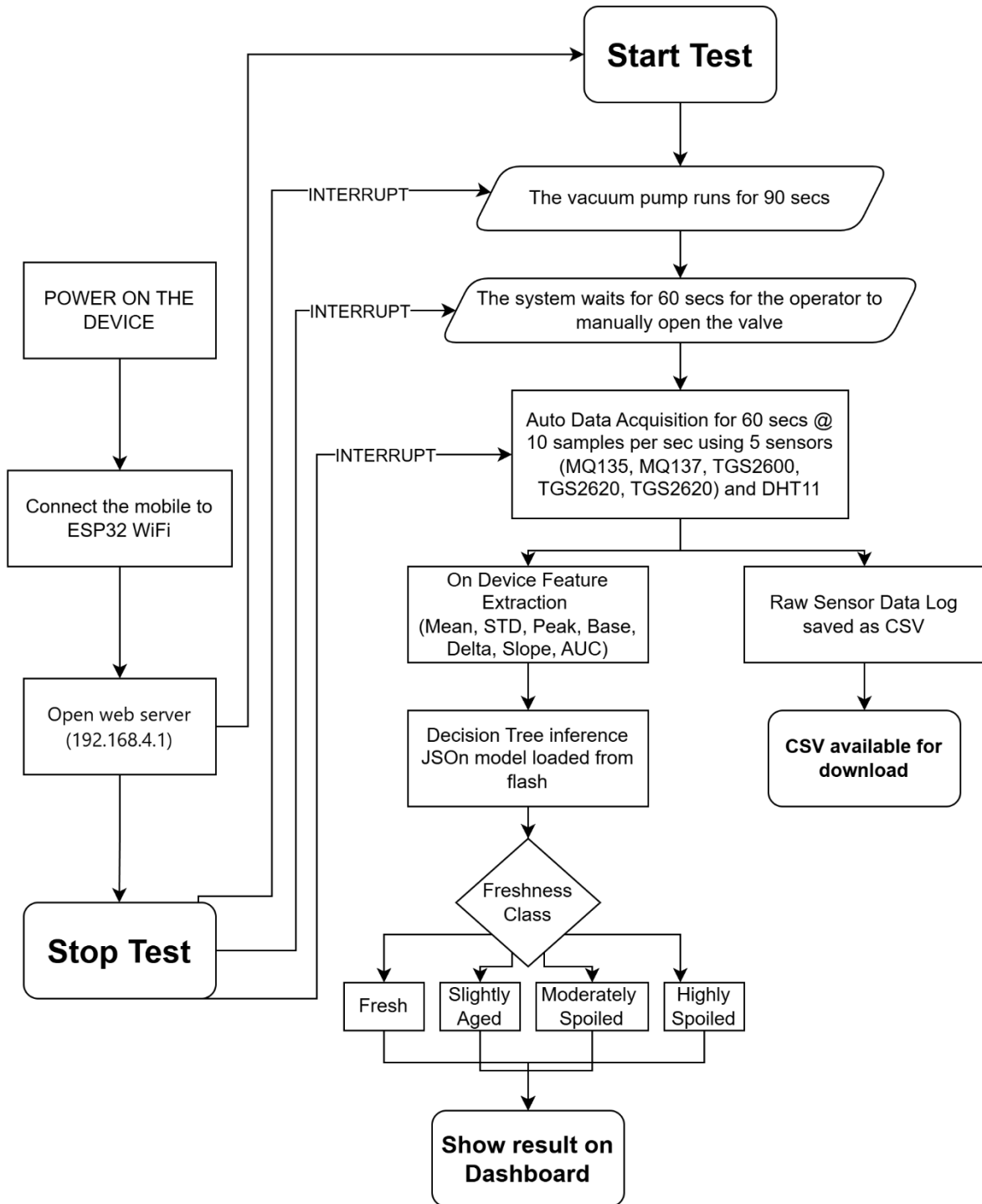


Figure 2. System architecture

4.3 Experimental Design and Data Collection

Training data were collected from three onions drawn from a single batch, stored at room temperature under identical conditions, and sampled at four time points corresponding to four freshness classes: Fresh (Day 0), Slightly Aged (Day 2), Moderately Spoiled (Day 4), and Highly Spoiled (Day 6). At each time point, five replicate measurements were taken per onion (3 onions $\times$ 4 classes $\times$ 5 replicates = 60 total observations), each comprising one full 60-second, 10 Hz acquisition window (approximately 600 samples per sensor per observation).

This protocol introduces a known confound: because each freshness class was sampled on a distinct calendar day, any variable correlated with the passage of time — ambient humidity, temperature drift, or sensor baseline drift — is statistically indistinguishable from the freshness label itself within this dataset. This limitation directly informed the deployable model design in Section 4.6 and is discussed further in Section 5.3.

4.4 Feature Engineering

Each 60-second acquisition window was reduced to a fixed-length feature vector per sensor channel rather than used in raw time-series form, both to yield a representation tractable for offline ensemble methods and on-device inference, and to summarize the chemically meaningful aspects of the sensor response (steady-state level, response magnitude, and recovery trend) rather than instantaneous noise. For each sensors, seven features were computed over the window:

- mean_s — arithmetic mean of all samples in the window
- std_s — sample standard deviation
- peak_s — maximum value observed
- base_s — mean of the first 10% of samples (pre-exposure baseline)
- $\text{delta}_s = \text{peak}_s - \text{base}_s$ — response amplitude
- auc_s — trapezoidal integral of the response (captures both magnitude and duration of the response)
- $\text{slope}_s = (\text{last sample} - \text{first sample}) / \text{window length}$ — net drift or recovery trend

This yields 35 sensor-derived features (5 sensors $\times$ 7 statistics) per observation, plus two environmental covariates (mean humidity, mean temperature) used only in the exploratory analysis of Section 4.5. The firmware computes all seven statistics incrementally — in a single pass over the incoming sample stream, with $O(1)$ memory per sensor — so the full ~ 600 -sample buffer is never retained in RAM (Section 4.7).

4.5 Phase 1: Offline Exploratory Analysis

Principal Component Analysis (PCA) was applied to the standardized 37-feature matrix (35 sensor-derived features plus 2 environmental covariates) to project the dataset into a reduced-dimensionality space and provide a preliminary, unsupervised assessment of how the four freshness classes are distributed relative to one another.

As an exploratory upper-bound benchmark, a Random Forest classifier (bagged ensemble of decision trees; MATLAB `fiscensembles`, Method = Bag) was trained on the full 37-feature set, including the two environmental covariates. Model performance was characterized using a stratified train-test holdout split together with k-fold cross-validation, and discriminative ability was further assessed via one-vs-rest receiver operating characteristic (ROC) analysis across the four freshness classes. Mean per-sensor response amplitude (δ_s) was additionally summarized by class to characterize the relative contribution of each gas sensor channel to class separation.

This benchmark was deliberately retained as an exploratory reference only: because the environmental covariates and sensor-derived features were collected under the day-correlated sampling protocol described in Section 4.3, any apparent discriminative power attributable to those covariates cannot be cleanly separated from the calendar-day confound. The Random Forest model was therefore excluded from consideration for on-device deployment, motivating the constrained model design described in Section 4.6.

4.6 Phase 2: Deployable Classifier Design

To obtain both an honestly validated accuracy estimate and a classifier suitable for embedded deployment, a second, independent training pass was performed under two deliberate constraints: (i) the model was restricted to a single shallow decision tree rather than an ensemble, keeping it representable as a small, human-auditable rule set within the ESP32's memory budget, and (ii) the two environmental covariates were excluded from the feature set entirely, to prevent the model from exploiting the day/class confound described in Section 4.3.

Tree depth was selected via k-fold cross-validation over a grid of candidate values for the maximum number of splits (`MaxNumSplits`), using the 35-feature sensor-only set. The configuration achieving the highest cross-validated accuracy was selected as the deployed model. Variable importance within the resulting tree structure was inspected to identify which sensor-derived features contributed to the final split structure, and which sensor channels, if any, did not contribute splits.

Model evaluation relied on out-of-fold (cross-validated) predictions to compute an honest confusion matrix and per-class precision, recall, and F1-score, rather than relying on training-set accuracy, which is inflated by overfitting given the small sample size.

4.7 Firmware Architecture for On-Device Inference

Two firmware-level design decisions enable the deployed model to run entirely on the ESP32 without continuous network dependency and without requiring recompilation each time the model is retrained.

First, feature extraction (Section 4.4) is implemented as a streaming accumulator: each of the five sensor channels maintains a running sum, sum-of-squares, running maximum, a fixed-size baseline accumulator (the first ~60 of the expected ~600 samples), first/last sample values, and a running trapezoidal integral, all updated in constant time per incoming sample. The seven statistical features are derived from these accumulators only once, at the end of the 60-second acquisition window, eliminating the need to buffer the full raw time series in memory.

Second, the decision tree itself is represented at runtime as a generic, data-driven node array (feature name and threshold, plus left/right child indices for internal nodes; a class index for

leaf nodes), rather than as compiled conditional logic. This structure is deserialized at boot from a JSON file held in onboard flash storage, or a built-in fallback model if none is present, and evaluated by a single generic tree-traversal routine. Retraining the model in MATLAB and exporting a new node array (Section 4.8) therefore changes only a data file on the device, not the compiled firmware.

4.8 Model Deployment and Update Mechanism

A purpose-built MATLAB export routine (`treeToJson.m`) serializes any trained `ClassificationTree` object into the node-array JSON format described in Section 4.7. The resulting file is transferred to the device over the same local Wi-Fi network used for routine operation, via an HTTP POST request to a dedicated `/update-model` endpoint exposed by the firmware; the device persists the received model to flash storage and reloads it immediately, without a reboot.

This mechanism was chosen in preference to (a) a cloud-inference architecture, which would introduce unnecessary latency, an internet-connectivity dependency, and hosting overhead for a model whose inference cost is negligible (a handful of threshold comparisons), and (b) full over-the-air firmware updates, which would require recompiling and re-flashing the entire program for what is, in practice, a change to a small set of numeric thresholds.

4.9 System Operation Workflow

In normal use, the device requires no connection to a laptop or external computer. An operator connects a Wi-Fi-enabled device (smartphone or laptop) to the ESP32 access point, initiates a test via the web dashboard, and the device autonomously executes the full evacuation–wait–acquisition–classification sequence (Section 4.2), reporting the result on both the web dashboard and the onboard OLED display, while archiving the raw sensor log for later retraining. A laptop running MATLAB is required only when the model itself is updated, not during routine testing.

RESULTS AND DISCUSSION

5.1 Phase 1: Offline Exploratory Analysis — Results

The PCA projection of the standardized 37-feature matrix (35 sensor-derived features plus 2 environmental covariates) showed that the first three principal components captured 76.3% of total variance (PC1: 54.7%, PC2: 12.5%, PC3: 9.1%; Figures 3–4). Visual inspection showed partial clustering with substantial inter-class overlap, consistent with the modest classification accuracies reported below.

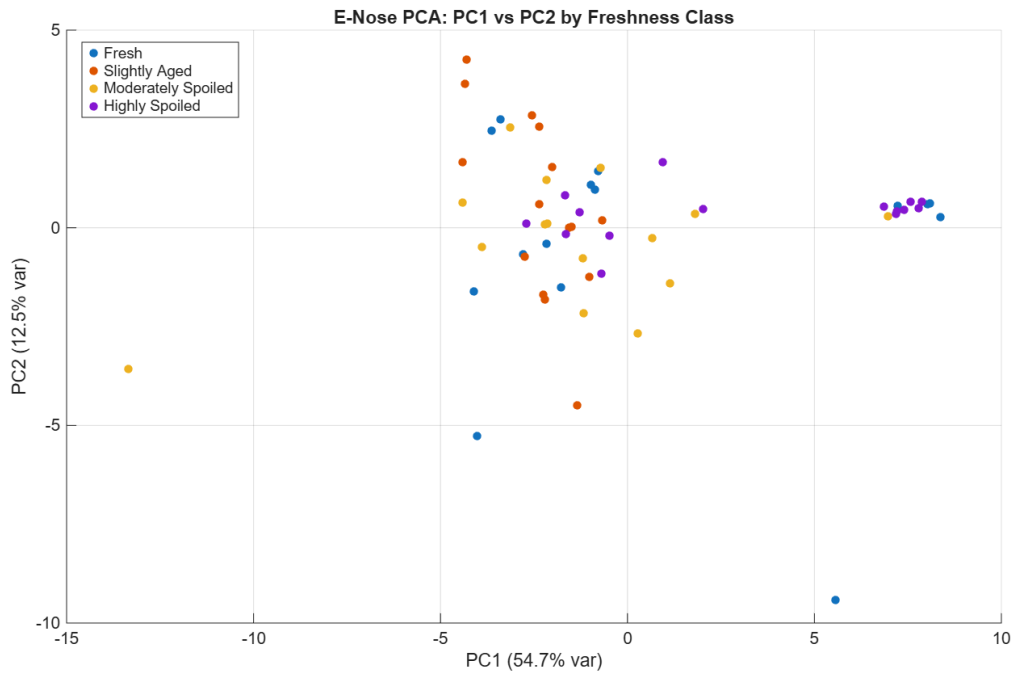


Figure 3. PCA projection onto the first two principal components, colored by freshness class.

The Random Forest exploratory benchmark, trained on the full 37-feature set, achieved 53.3% holdout accuracy on a stratified 75/25 train-test split; 5-fold cross-validation gave 75.0% accuracy and a macro-averaged one-vs-rest AUC of 0.903 (Figures 5–6).

Inspection of the trained ensemble showed that the environmental covariates (mean humidity, mean temperature) ranked among the most important predictors. Because these covariates are perfectly collinear with the day of measurement (Section 4.3), this indicates that part of the Random Forest's apparent accuracy reflects discrimination between data-collection days rather than discrimination between true freshness states. This benchmark is therefore reported as an exploratory upper-bound reference only and was not selected for deployment.

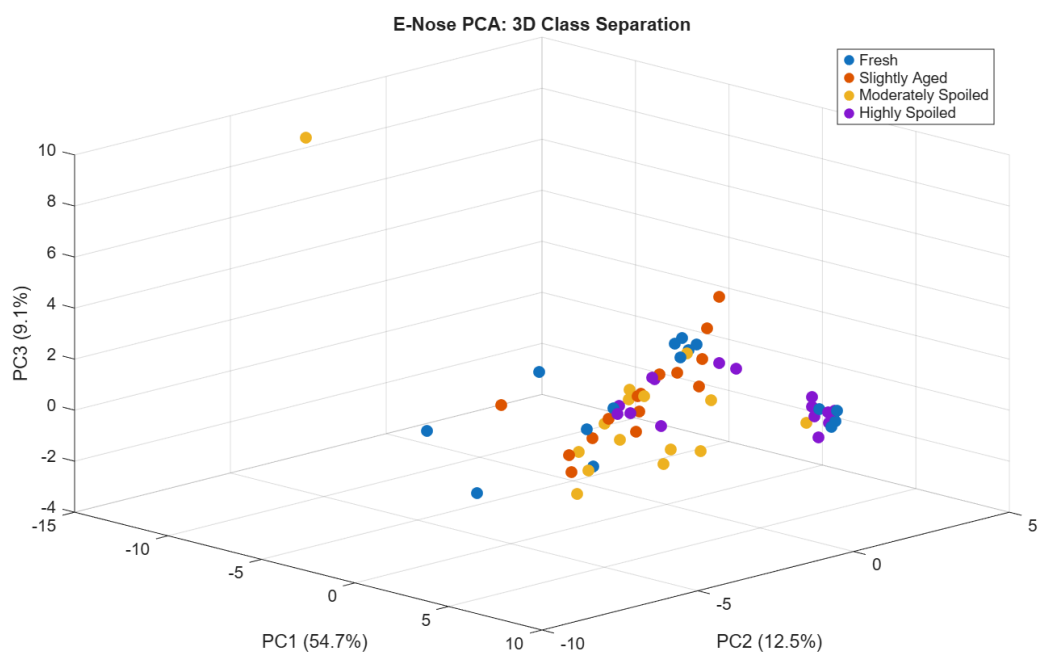


Figure 4. PCA projection onto the first three principal components.

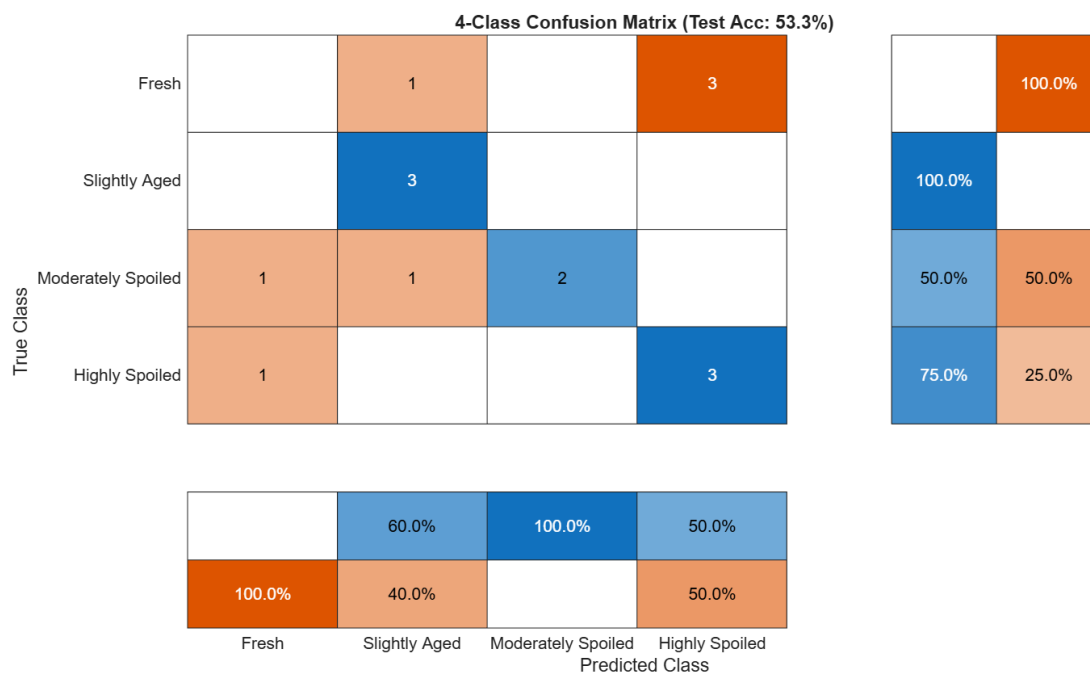


Figure 5. Random Forest holdout confusion matrix (37-feature set, including environmental covariates).

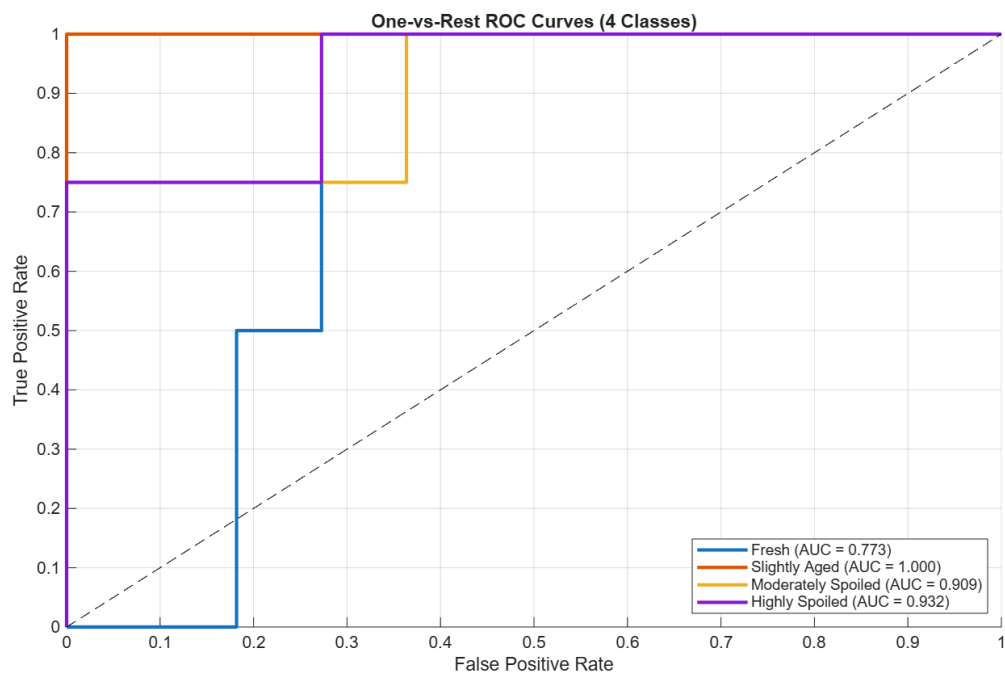


Figure 6. One-vs-rest ROC curves for the Random Forest benchmark, by class.

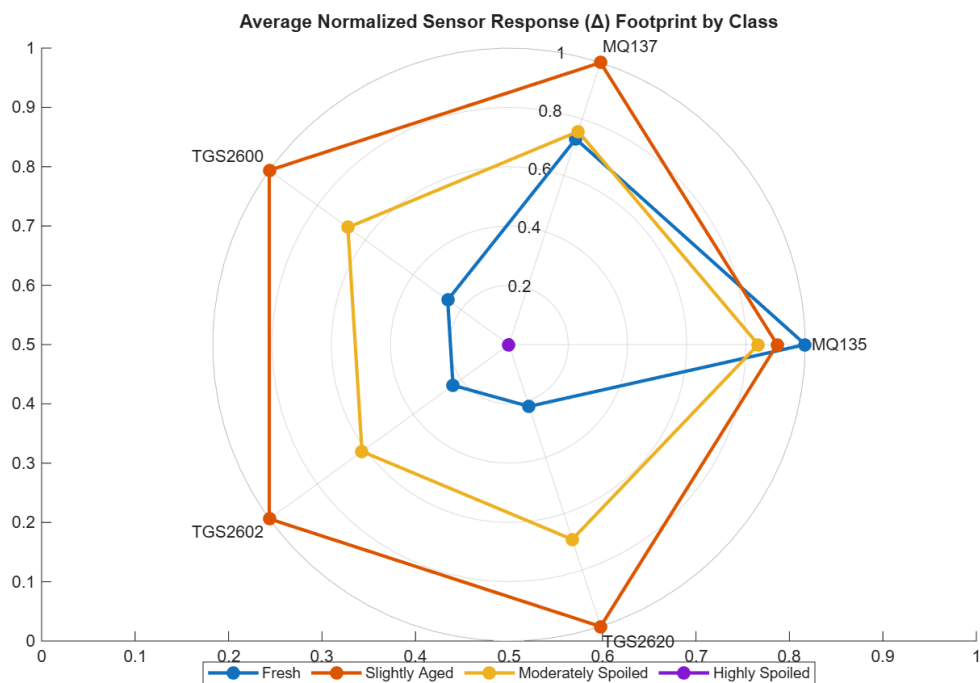


Figure 7. Mean per-sensor response amplitude (Δ) by freshness class, normalized per sensor.

A radar plot of the mean response amplitude (δa_s) per class (Figure 7) is included for exploratory completeness. Highly Spoiled exhibited the lowest mean amplitude on every sensor channel in this dataset — an inversion of the spoilage-VOC relationship typically expected in the e-nose literature — which is most parsimoniously explained by the day/humidity confound described above rather than by genuinely reduced odor intensity in advanced spoilage.

5.2 Phase 2: Deployable Classifier — Results

Tree depth was evaluated via 5-fold cross-validation over $\text{MaxNumSplits} \in \{3, 7, 15, 31\}$ (approximate maximum depths 2–5), using the 35-feature sensor-only set (Table 3).

MaxNumSplits	Approx. Max Depth	5-Fold CV Accuracy
3	2	50.0%
7	3	65.0%
15	4	56.7%
31	5	53.3%

Table 3. Cross-validated accuracy by maximum tree complexity (sensor-only feature set).

$\text{MaxNumSplits} = 7$ (effective depth 3) achieved the highest cross-validated accuracy and was selected as the deployed model (Figure 8). The resulting tree selects almost exclusively on MQ135- and MQ137-derived features (MQ137_mean, MQ135_std, MQ135_peak, MQ135_slope, MQ137_peak); the remaining three sensor channels (TGS2600, TGS2602, TGS2620) contributed no splits in the final structure.

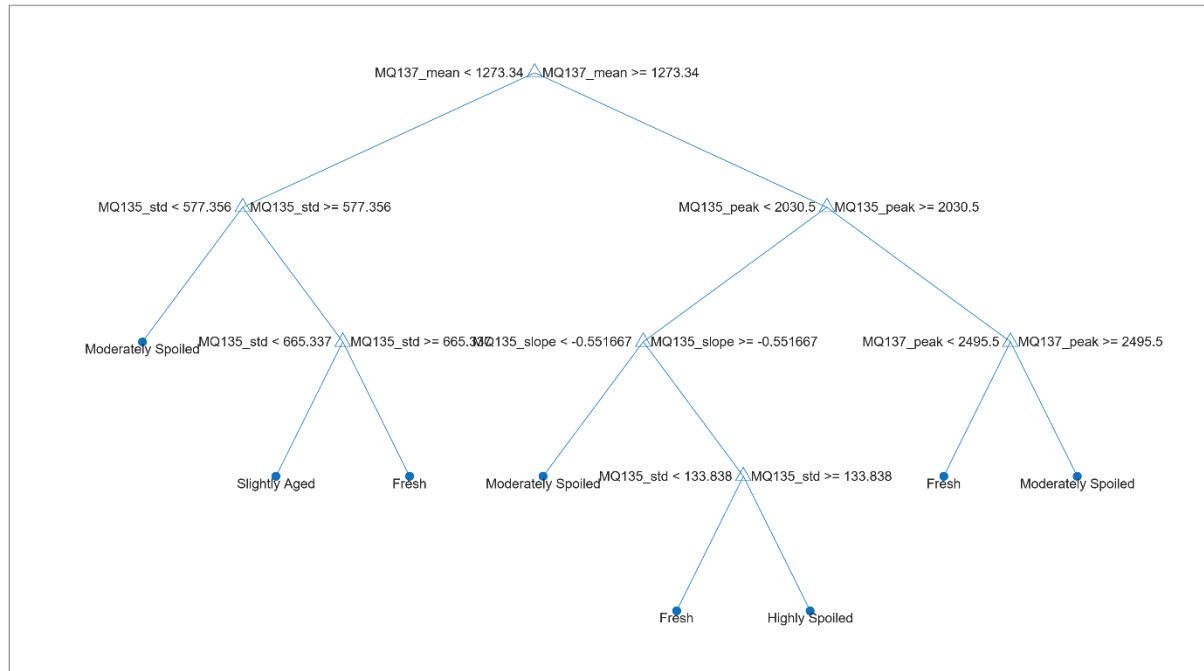


Figure 8. Structure of the deployed decision tree (sensor-only features, $\text{MaxNumSplits} = 7$).

Out-of-fold (5-fold cross-validated) predictions were used to compute an honest confusion matrix and per-class precision, recall, and F1-score (Figure 9, Table 4), rather than relying on training-set accuracy (83.3%), which is inflated by overfitting given the small sample size.

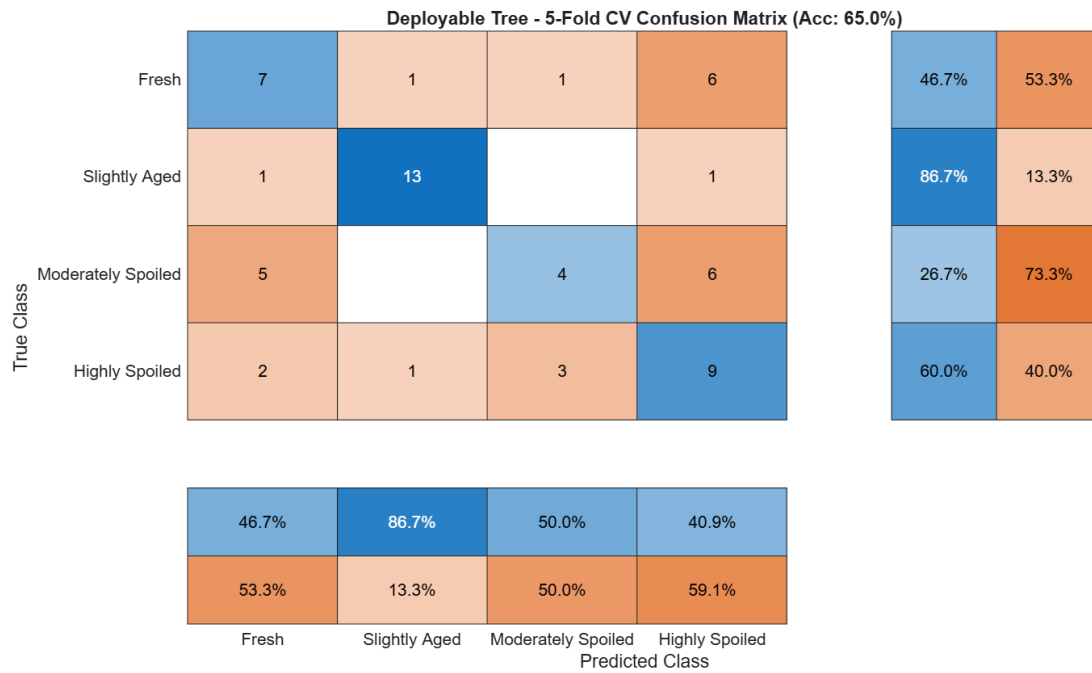


Figure 9. 5-fold cross-validated confusion matrix for the deployed decision tree.

Class	Precision	Recall	F1-score
Fresh	0.467	0.467	0.467
Slightly Aged	0.867	0.867	0.867
Moderately Spoiled	0.500	0.267	0.348
Highly Spoiled	0.409	0.600	0.486

Table 4. Per-class classification performance, 5-fold cross-validated. Overall accuracy: 65.0%. Macro-averaged F1-score: 0.542.

Slightly Aged was the most reliably identified class ($F1 = 0.867$); Moderately Spoiled was the weakest ($F1 = 0.348$), most frequently confused with both its neighboring classes.

5.3 Limitations

The principal limitation of the present dataset is the confound between freshness class and calendar day of measurement (Section 4.3): because each class was sampled on a single, distinct day, day-correlated environmental or electronic drift cannot be statistically separated from true spoilage-related sensor response. The reported Phase 2 accuracy (65.0% cross-validated, macro-F1 = 0.542) should be interpreted as a conservative estimate obtained after excluding the two most obviously day-correlated covariates (humidity, temperature), rather than as a guarantee that all day-correlated influence has been removed from the remaining 35 sensor-derived features.

A second limitation is the small sample size ($n = 60$; 3 biological replicates per class), which limits the statistical power of the cross-validation estimates and increases the risk that the selected tree structure reflects sampling idiosyncrasies rather than generalizable spoilage chemistry.

A third limitation, identified directly from the structure of the deployed model, is that the selected decision tree relies almost exclusively on the MQ135 and MQ137 channels; the three TGS-series sensors, while logged and available, did not contribute to the final classifier, and the system's robustness to single-sensor drift or failure has therefore not been empirically validated.

Future data collection should interleave or counterbalance freshness classes across calendar days, rather than assigning one class per day, to break the confound described above, and should expand the sample size per class to support more reliable cross-validation and a more balanced reliance across all five sensor channels.

CONCLUSION AND FUTURE SCOPE

6.1 Final Summary

This internship project set out to design, fabricate, and characterise a low-cost, standalone electronic nose (E-nose) capable of non-destructive, real-time freshness classification of onion bulbs, addressing a documented gap in postharvest quality monitoring for Indian agricultural supply chains. Over the course of the two-month internship period, the work progressed from sensor selection and literature grounding through to a fully functional hardware-firmware-analysis pipeline.

The completed system integrates a five-sensor metal-oxide-semiconductor (MOS) gas sensor array (MQ135, MQ137, TGS2600, TGS2602, TGS2620) and a DHT11 temperature-humidity sensor around an ESP32 microcontroller, housed in a custom acrylic chamber assembly fabricated in-house. The device executes a fully automated four-stage measurement protocol — chamber evacuation, manual valve-opened headspace admission, synchronised 10 Hz data acquisition, and on-device classification — controllable entirely through a self-hosted Wi-Fi web dashboard, with no dependency on external computing infrastructure during routine operation. A parallel offline pipeline, developed in MATLAB, was used to extract chemically meaningful features from the raw sensor logs and to train, validate, and export a deployable classification model back onto the device. In this sense, the core objective of the internship — building an end-to-end, edge-deployable E-nose pipeline for onion freshness assessment, from raw VOC sensing to displayed classification result — was substantively achieved within the available time, with machine learning classification successfully transitioned from a planned future phase (as stated in the Introduction) to a completed, evaluated component of the system.

6.2 Key Findings & Data Insights

Several technical findings emerged from the exploratory and deployable-model analyses that carry meaningful implications for the design of E-nose systems in this domain:

- **Dimensionality reduction confirmed partial but incomplete class separability.** PCA on the standardised 37-feature matrix showed that the first three principal components captured 76.3% of total variance, but visual inspection revealed substantial inter-class overlap among the four freshness states (Fresh, Slightly Aged, Moderately Spoiled, Highly Spoiled), foreshadowing the moderate classification accuracies obtained downstream.
- **Environmental covariates introduced a day-correlated confound that inflated apparent accuracy.** The exploratory Random Forest benchmark, trained on the full feature set including humidity and temperature, achieved a cross-validated accuracy of 75.0% (AUC = 0.903) — but inspection of feature importances showed the environmental covariates ranking among the most influential predictors. Because each freshness class was sampled on a single, distinct calendar day, this indicates the model was partly discriminating between *data-collection days* rather than genuine spoilage states. This is among the most important methodological findings of the project, as it

directly motivated the exclusion of environmental covariates from the deployable model.

- **A constrained, sensor-only decision tree provided an honest, deployment-ready accuracy estimate.** Restricting the feature set to the 35 sensor-derived statistics and selecting tree depth via cross-validation (MaxNumSplits = 7) yielded a deployable classifier with 65.0% cross-validated accuracy and a macro-averaged F1-score of 0.542 — a more defensible figure than the inflated Random Forest benchmark, precisely because it cannot exploit the day/class confound.
- **Class-wise performance was uneven, with Slightly Aged most reliably detected (F1 = 0.867) and Moderately Spoiled the weakest (F1 = 0.348),** frequently confused with both its neighbouring classes — consistent with the expectation that intermediate spoilage states present a less distinct VOC signature than the two extremes.
- **Sensor-level analysis revealed an unexpected reliance on two channels.** The deployed decision tree split almost exclusively on MQ135- and MQ137-derived features; the three TGS-series sensors, despite being logged throughout, contributed no splits to the final model. The radar-plot analysis additionally showed an inversion of the expected spoilage-VOC relationship (Highly Spoiled exhibiting the lowest mean response amplitude across all channels), most plausibly attributable to the same day/humidity confound rather than to genuinely diminished odour intensity at advanced spoilage.
- **Cost-efficiency was demonstrated alongside technical performance.** The complete hardware prototype was fabricated for approximately ₹5,960 (~\$72 USD) — substantially below both prior published MOS-based E-nose systems (~\$300) and commercial E-nose instruments (₹5–50 lakhs), reinforcing the practical relevance of the system for resource-constrained agricultural settings.

6.3 Deliverables & Contributions

This internship produced a set of concrete, reusable deliverables for the host laboratory (CHCI Lab, SCEE, IIT Mandi) and for the continuation of this research line:

- **A fabricated, functioning hardware prototype** — comprising the custom acrylic sensor and sample chambers, the integrated five-sensor MOS array with DHT11, vacuum pump and valve gas-delivery subsystem, and OLED status display — ready for continued data collection or demonstration.
- **Production firmware for the ESP32**, implementing the automated evacuation–wait–acquisition–classification measurement cycle, an onboard Wi-Fi access point and web dashboard for control and result display, CSV logging to flash storage, and a JSON-based, over-the-air model update mechanism that allows the deployed classifier to be retrained and redeployed without firmware recompilation.
- **A validated MATLAB analysis pipeline**, covering feature extraction, exploratory PCA and Random Forest benchmarking, cross-validated decision-tree training and selection, and a model-export routine (treeToJSON.m) bridging offline training to on-

device inference.

- **An annotated, labelled dataset** of 60 measurement observations across four freshness classes, together with the documented sample-preparation and spoilage-induction protocol used to generate it — providing a foundation that future data-collection efforts can directly extend.
- **A consolidated literature review** of 19–20 reference works on E-nose-based onion and potato quality assessment, synthesising prior sensor-array designs, baseline-correction and feature-extraction practices, and classifier choices (notably SVM/LDA performance benchmarks), which directly informed design decisions in this work and will continue to inform the planned conference paper.
- Collectively, these outputs advance the host group's broader interest in low-cost, IoT-enabled E-nose systems for postharvest quality monitoring, and identify a concrete, previously underexplored gap — the integration of real-time wireless connectivity with edge-deployable inference — that this prototype demonstrates is practically achievable.

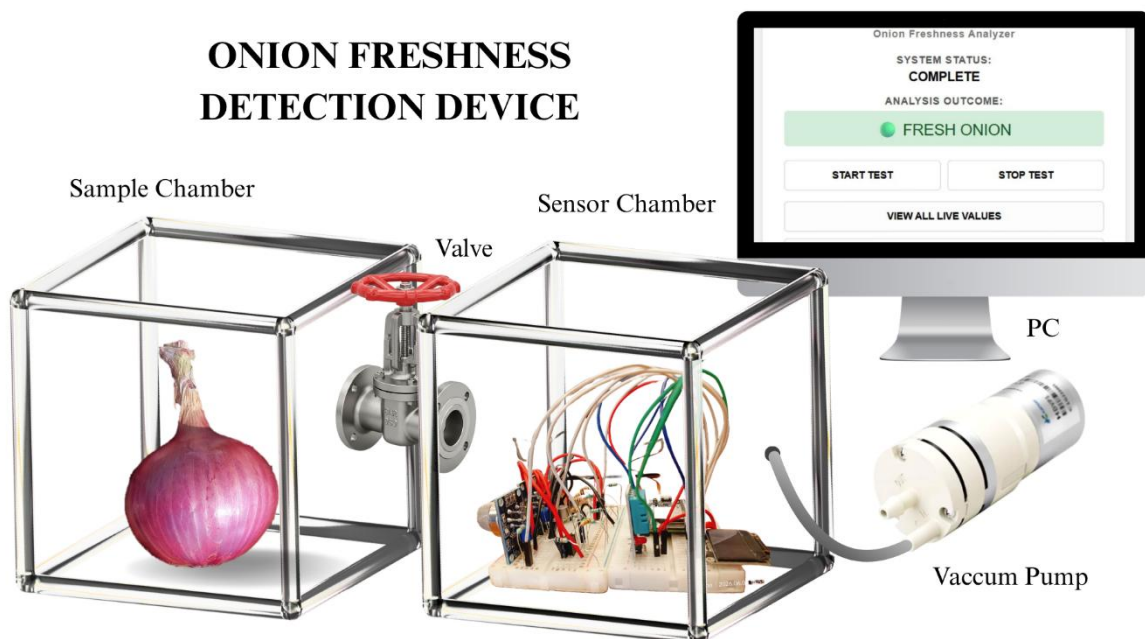


Figure 10. The Hardware Prototype

6.4 Future Work

Building on the foundation established during this internship, the following directions are recommended as the most logical and high-value next steps:

1. **Decouple the freshness class from calendar day in the data-collection protocol.** The single largest limitation identified in this work is that each freshness class was sampled on its own distinct day, confounding spoilage-related signal with day-correlated environmental drift. Future data collection should interleave or counterbalance classes across days (e.g., sampling onions from all four freshness states within each collection session, drawn from a staggered initial cohort) to statistically separate true spoilage

chemistry from incidental temporal drift, and should expand the dataset well beyond the present $n = 60$ (3 biological replicates per class) to support more statistically robust cross-validation and reduce the risk of the classifier reflecting sampling idiosyncrasies.

2. **Investigate and address the TGS-sensor underutilisation.** Since the deployed model relies almost exclusively on the MQ135 and MQ137 channels, a focused follow-up should examine why the TGS2600/2602/2620 sensors contributed no useful splits — whether through sensor-specific recalibration, exploration of alternative feature definitions (e.g., the relative-response and relative-baseline-correction approach identified as best practice in the literature review), or targeted experiments isolating each sensor's response to known onion-relevant VOCs. This would both improve overall accuracy and increase the system's robustness to single-sensor drift or failure, which the present model has not been validated against.
3. **Extend the system toward the originally envisioned conference paper and explore classifier alternatives.** With the deployable pipeline now operational, the next phase should formalise the dataset and findings into the planned conference submission, while also benchmarking alternative classifiers (SVM with RBF kernel, k-NN, or shallow neural networks) against the current decision-tree baseline now that a larger, deconfounded dataset is available — leveraging the prior literature's finding that SVM has outperformed LDA in comparable onion-classification tasks. Longer-term, the JSON-based model-update mechanism already built into the firmware positions the system well for iterative refinement as new data and classifier variants become available, without requiring further hardware or firmware redevelopment.

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